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Chatbot to chat with medical books using Retrieval-Augmented Generation Model

Project Name: MedBuddy: Medical Textbook AI Assistant

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Abstract

This research paper presents a novel AI tool aimed at revolutionizing medical education by addressing the specific needs of medical students, fostering effective learning, and managing stress. The implementationleverages cutting-edge technology known as Large Language Models (LLMs), with a focus on utilizing Gemini, Streamlit, and Trulens. The digitalassistant serves as a friendly and accurate resource, responding promptly to complex medical queries, thereby optimizing study time. The study meticulously examines the balance between technical innovation and practical benefits, emphasizing the project's goal of not only creating a sophisticated tool but also providing tangible assistance to medical students. The report outlines the successful development and deploy-ment of the AI-powered study buddy, showcasing its potential to significantly improve learning environments, particularly indemanding fields such as medicine. Notably, the project empha-sizes the importance of groundedness in its chatbot, achieving an impressive accuracy rate of approximately 84% for a single textbook. The integration of Streamlit, Gemini, and Trulens contributes to creating a seamless and user-friendly experience for medical students.
This research underscores the broader implications of AI in education, demonstrating how advancements in technology can positively impact learning outcomes. By reducing the timespent on information retrieval, the AI study buddy empowers medical students to concentrate more on core learning objectives, ultimately facilitating their journey to becoming proficient and successful medical practitioners.

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Introduction

- The rising demand for skilled professionals necessitates better preparation tools. To bridge the gap in resources and empower individuals, our project develops an AI-powered platform.
- This application is designed to tailor questions and insights to the unique requirements of the user's focus area.
- The platform offers valuable feedback on user responses and the correct answers of the questions based strictly on provided documentation.
- By offering a comprehensive and personalized preparation experience, we help users refine their skills, boost their confidence, and ultimately succeed in their academic and career pursuits.



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Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	AI Based Medical Data Retrieval System[2024]	Ayesha Sayyad Sayyad	Uses AI and NLP to assess candidate responses	General AI lacks domain-specific medical grounding and factual accuracy.
2	Role of ChatGPT and AI in Medical Education [2023]	Gunther Eysenbach	Investigated standard LLMs in medical learning	High risk of authenticity issues and "hallucinations" without a retrieval system.
3	AI Chatbots in Public Health Education [2023]	Francesco Baglivo et al.	Evaluated AI chatbots for vaccination training	Lacked professional verification and reliable source-tracking for answers.
4	Ontology Based Personal Assistive Application [2023]	N. V. Rohil et al.	Uses ontology modeling and semantic reasoning for health	Highly complex to update with new medical textbooks and literature.

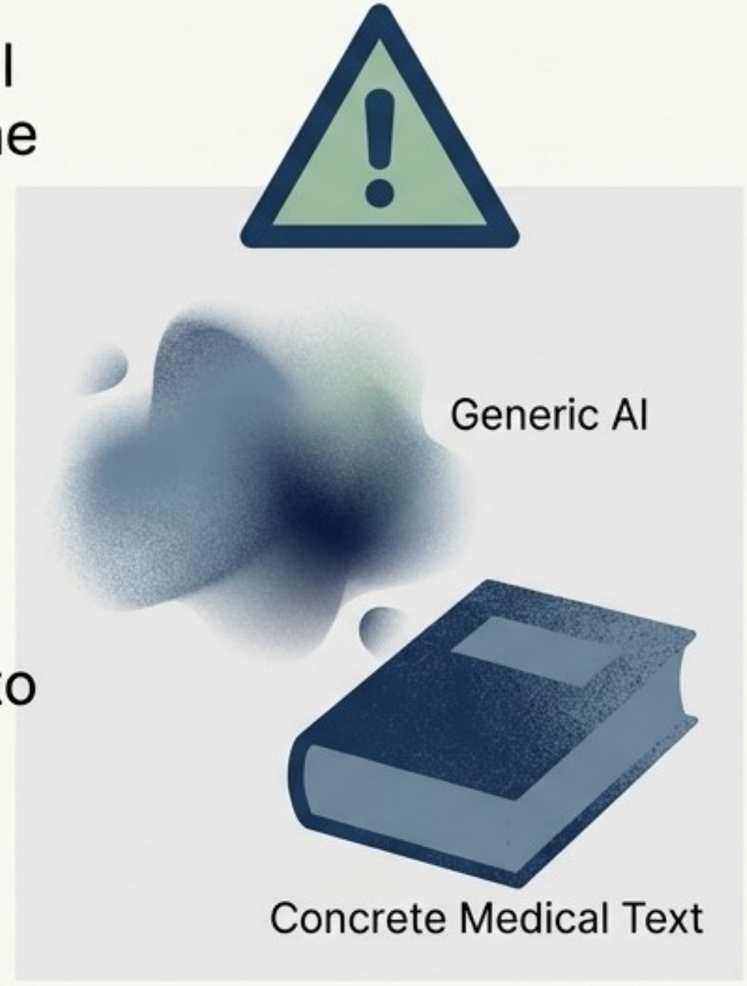
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Problem Identification

- **Lack of Transparency:** Information provided by normal generic AI is neither transparent nor concrete about the source of the information.
- **The Hallucination Risk:** Generic models use LLMs to generate answers which have a high possibility of hallucination and cannot provide a particular source for those answers.
- **Critical Consequences:** Students become vulnerable to false information or data not related to their syllabus. Misleading medical students with wrong information is a critical failure that might cost lives in the future.

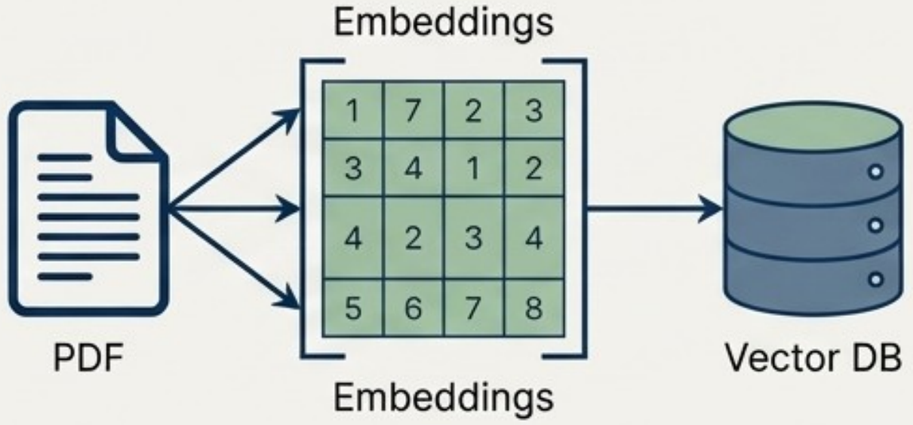


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Proposed System

MedBuddy Overview: Uses Google Gemini as the core AI for the purpose of Llama indexing to ensure accuracy in the data retrieval of uploaded PDFs.

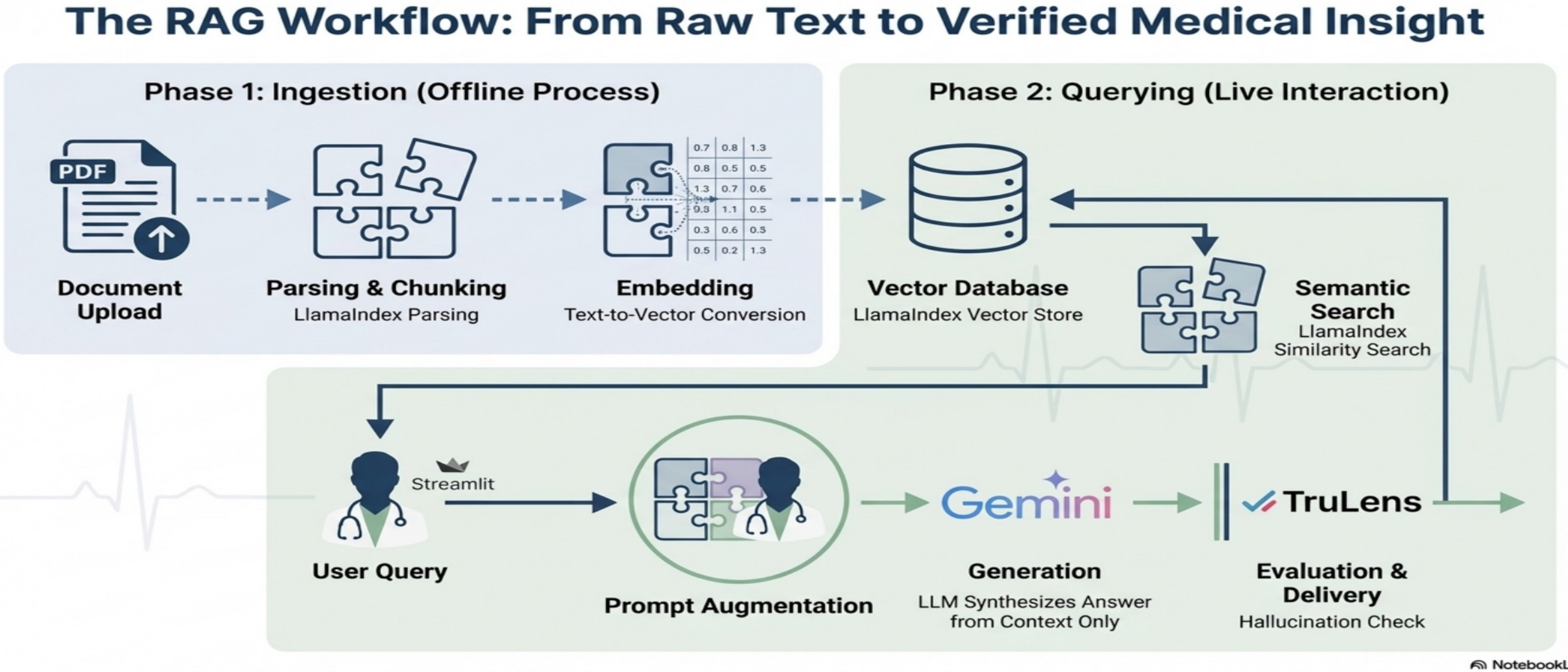


Key Concepts:

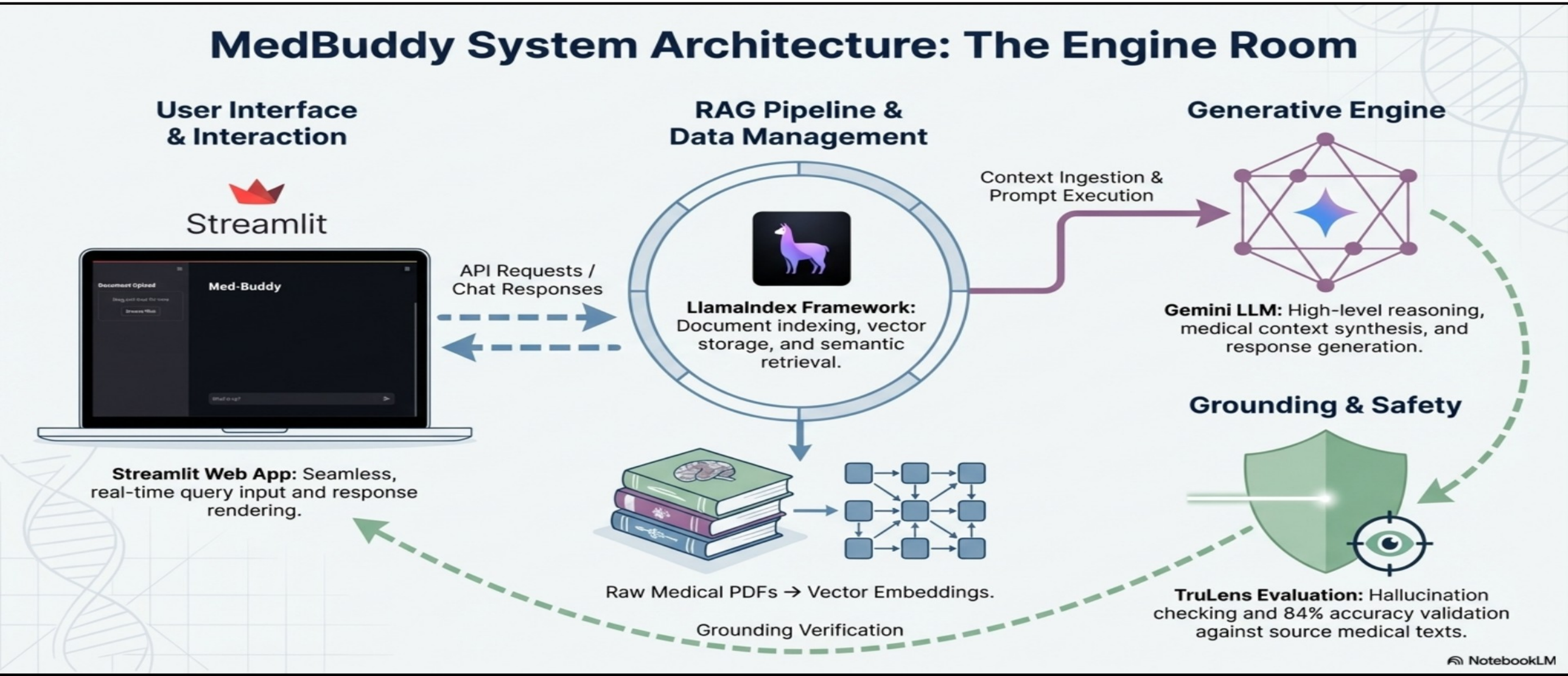
- **Embeddings**: The process of turning human words and sentences into a list of numbers so the computer can mathematically understand their meaning.
- **Vector Database**: A special storage space where we keep these ‘number-words,’ allowing the AI to instantly find paragraphs that match the user's question.
- **LLM Grounding:** A strict rule placed on the AI, forcing it to only generate an answer using the text provided, completely stopping it from making up outside facts.

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Workflow


The RAG Workflow: From Raw Text to Verified Medical Insight
Phase 1: Ingestion (Offline Process)
PDF
Document Upload
Parsing & Chunking
LlamaIndex Parsing
0.7 0.8 1.3
0.8 0.5 0.5
1.3 0.7 0.6
0.3 1.1 0.5
0.3 0.6 0.5
0.5 0.2 1.3
Embedding
Text-to-Vector Conversion
Phase 2: Querying (Live Interaction)
Vector Database
LlamaIndex Vector Store
Semantic Search
LlamaIndex Similarity Search
Streamlit
User Query
Prompt Augmentation
Gemini
Generation
LLM Synthesizes Answer from Context Only
TruLens
Evaluation & Delivery
Hallucination Check
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System Architecture



Software and Hardware Requirements

Software

- Pyhton (Core programming language)
- LlamaIndex (Framework to connect data to LLMs)
- Trulens (Library to evaluate Ai groundness)
- Google Gemini API
- Streamlit (Creates the web application UI)

Hardware

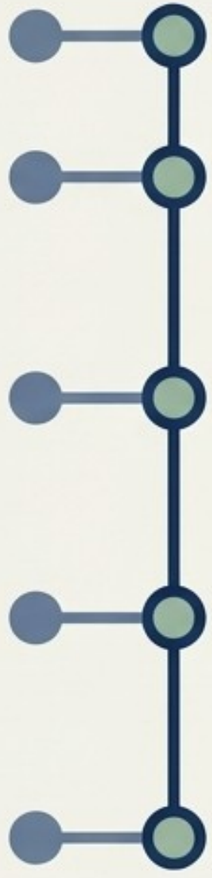
- Processor - Intel i5 or i7, AMD Ryzen 5 or 7
- Memory - min 8GB RAM
- Storage - SSD(solid state drive) recommended, min 256GB
- Operating System – 64 bit version of Windows 10/11, macOS or Linux

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Project Plan

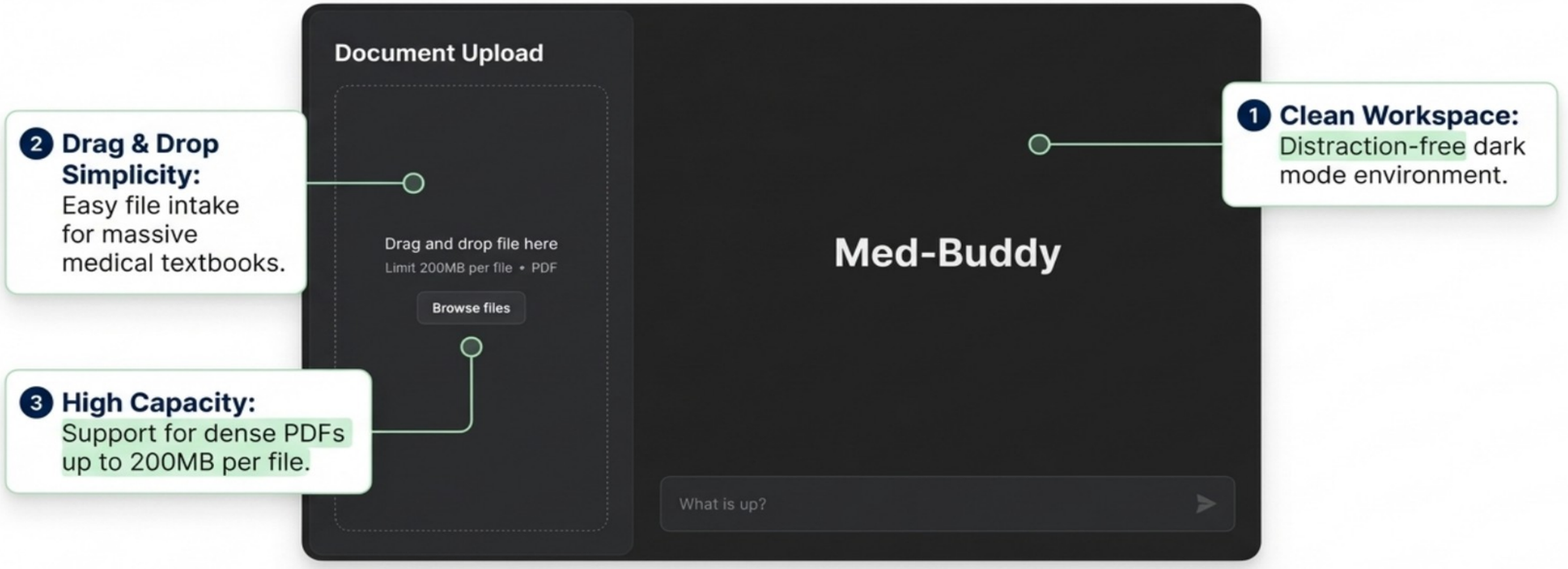
- Jan 16, 2026: Initial initiation and UI set-up
- Feb 1, 2026: Building Streamlit interface and configuring Python libraries
- Feb 15, 2026: Data processing module implementing PDF chunking and embedding
- Mar 1, 2026: Integrating of the LlamaIndex vector database and the Gemini API
- Mar 6, 2026: 50% project completion for first evaluation



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Current Work

A frictionless interface for secure medical document upload

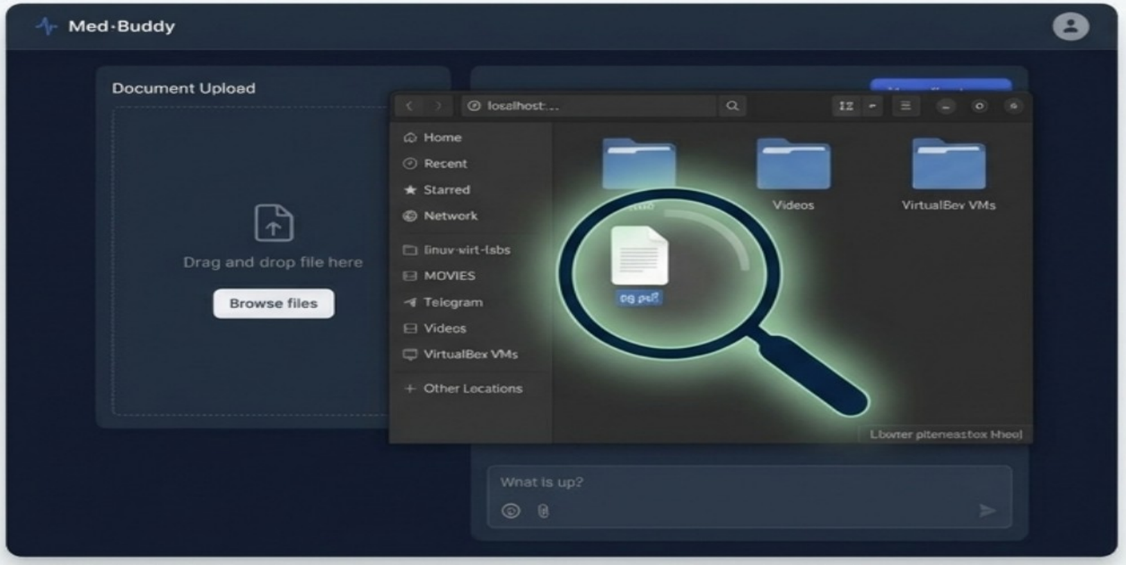


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Current Work

Seamless bridging between local environments and the web application



Users can instantly map local medical literature (e.g., standard curriculum PDFs) **directly** into the secure Med-Buddy indexing pipeline **without cloud exposure risks.**

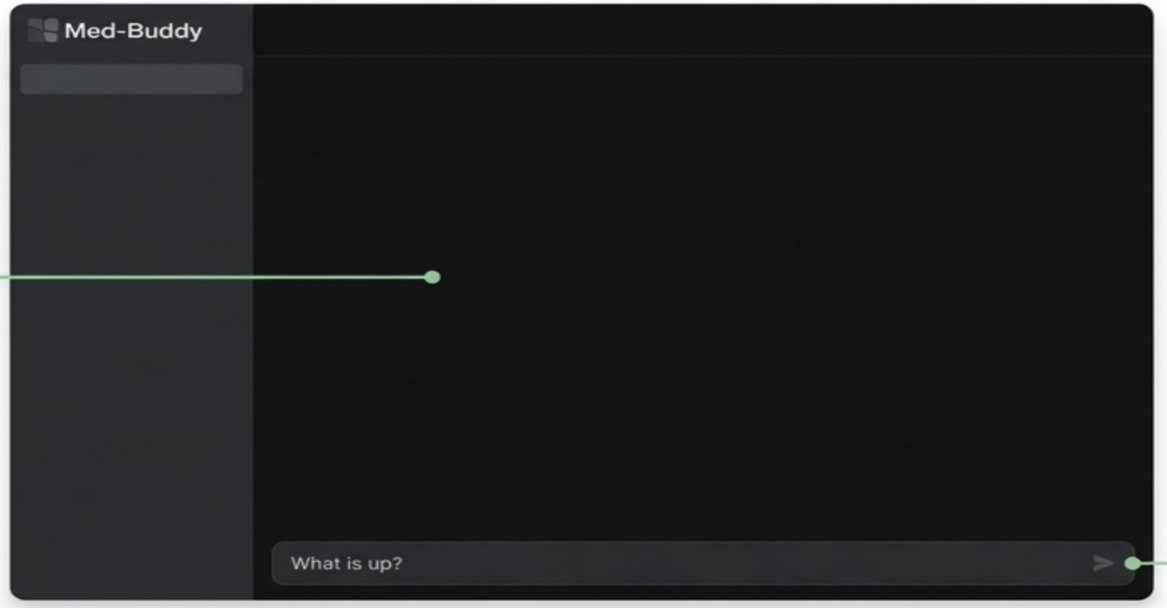
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Current Work

A focused chat environment restricted strictly to verified contexts

1 The Learning Zone:
Once uploaded, the Gemini engine is locked to the document context.



2 Instant Retrieval:
Queries entered here bypass generic web data, fetching only mathematically matched textbook chunks.

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Conclusion

Med-Buddy successfully provides a safe, grounded, and hallucination-free Ai assistant. By combining Google Gemini with a rigorous RAG pipeline, it achieves high accuracy. It democratizes access to complex medical knowledge, significantly reducing the mental burden on students and allowing them to study smarter, not harder.

Future Scope

Phase	Action Item	Priority
Evaluation	Measure groundedness and context relevance using TruLens	High
Optimization	Refine chunk size and memory usage for large PDF textbooks	Medium
Deployment	Finalize system for local Streamlit execution and final presentation readiness	High

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References

1. **Peri, S. et al.**, "Chatbot to chat with medical books using Retrieval-Augmented Generation Model," *IEEE International Conference*, 2024.
2. **N. V. Rohil and A. Nair**, "Ontology Based Personal Assistive Application," *4th IEEE Global Conference for Advancement in Technology (GCAT)*, 2023.
3. **P. C. Nair et al.**, "HR based Chatbot using Deep Neural Network," *International Conference on Inventive Computation Technologies (ICICT)*, 2022.
4. **R. S. Nambiar and D. Gupta**, "Dedicated Farm-Haystack Question Answering System for Pregnant Women," *12th International Conference on Cloud Computing*, 2022.
5. **P. C. Nair et al.**, "HR based Chatbot using Deep Neural Network," *International Conference on Inventive Computation Technologies (ICICT)*, 2022.